Calculation Workbook — VBF Power-Tool Cell (V4)

Input Parameters

Parameter	Value	Unit	Source
Positive electrode thickness	4.2e-05	m	params_r2_v4.json
Negative electrode thickness	5.8e-05	m	params_r2_v4.json
Positive electrode porosity	0.4	-	params_r2_v4.json
Negative electrode porosity	0.38	-	params_r2_v4.json
Positive particle radius	8e-07	m	params_r2_v4.json
Negative particle radius	1e-06	m	params_r2_v4.json
Electrolyte conductivity	2.4	S/m	params_r2_v4.json (design override)
Electrolyte diffusivity	3.5e-10	m2/s	params_r2_v4.json (design override)
Cation transference number	0.5	-	params_r2_v4.json (design override)
Total heat transfer coefficient	100	W/m2/K	params_r2_v4.json (forced-air cooling)
Electrode height	0.065	m	parameter set
Electrode width	1.58	m	parameter set
Positive solid density	3262	kg/m3	parameter set
Negative solid density	1657	kg/m3	parameter set
Positive CC density (Al)	2700	kg/m3	parameter set
Negative CC density (Cu)	8960	kg/m3	parameter set
Electrolyte density (estimate)	1.2	g/cm3	literature estimate, annotated

Capacity and Energy

Quantity	Value	Unit	Formula / source
1C discharge capacity	3.437599389317619	Ah	run-pyamm 1C_discharge, DFN, 25 C (r2_v4_1c_dfn.json)
5C discharge capacity	3.392723700476792	Ah	run-pyamm 5C_discharge, DFN, 25 C (r2_v4_5c_dfn.json)
5C capacity retention	0.9869456315996917	-	5C / 1C
4C CC-phase charge accepted	0.6172579731035653	Ah	run-pyamm 4C_charge_45C, DFN, to 4.2 V cutoff
Energy (1C full discharge)	12.42340893099891	Wh	integral V*I dt over 1C discharge
Midpoint voltage	3.881233322093622	V	voltage at 50% DOD
DCR (1C, first 10% time)	0.001257286775024369	ohm	(V@t0 - V@10%time)/I_1C

Energy Density and Power

Quantity	Value	Unit	Formula / source
Cell mass (contract layer mass)	30.299893208	g	sum(layer thickness x (1-porosity) x density x area); electrolyte excluded
Stack thickness	140	um	sum of layer thicknesses
Electrode area	0.1027	m2	height x width
Gravimetric energy density	410.0149411654953	Wh/kg	energy / mass (contract caliber)

Quantity	Value	Unit	Formula / source
Volumetric energy density	864.0568181248374	Wh/L	energy / (thickness x area)
Power density (theoretical peak)	111807.4344838161	W/kg	V_OC^2 / (4 x DCR) / mass
Layer mass positive	0.0822024	kg/m2	calc-energy output
Layer mass negative	0.05958572	kg/m2	calc-energy output
Layer mass pos CC	0.0432	kg/m2	calc-energy output
Layer mass neg CC	0.10752	kg/m2	calc-energy output
Layer mass separator	0.00252492	kg/m2	calc-energy output

N-P and Process

Quantity	Value	Unit	Formula / source
Positive capacity density	63104	Ah/m3	parameter set (per m3 active material)
Negative capacity density	33133	Ah/m3	parameter set (per m3 active material)
Positive max stoichiometry	0.6	-	parameter set
Negative max stoichiometry	0.62	-	parameter set
N/P ratio (full-range capacity density)	0.75	-	neg cap dens x thick / pos cap dens x thick; see design_spec
Positive areal density	82.2	g/m2	thickness x (1-porosity) x density
Negative areal density	59.6	g/m2	thickness x (1-porosity) x density
Positive compaction density	1957	kg/m3	density x (1-porosity)
Negative compaction density	1027	kg/m3	density x (1-porosity)
Pore volume	4.57	mL	sum(porosity x thickness x area) incl. separator
Electrolyte fill (estimate)	5.5	g	pore volume x 1.2 g/cm3 (literature density, annotated)